

Creating Microservices for account and loan

Account Services:

Spring Initializer:

The screenshot shows the Spring Initializr web application interface. The header includes the Spring logo and the text "spring initializr". On the left, there is a sidebar with a hamburger menu icon and a circular arrow icon. The main content area is divided into several sections:

- Project:** Radio buttons for "Gradle - Groovy", "Gradle - Kotlin", and "Maven" (selected).
- Language:** Radio buttons for "Java" (selected), "Kotlin", and "Groovy".
- Spring Boot:** Radio buttons for "4.1.1 (SNAPSHOT)", "4.1.0" (selected), "4.0.8 (SNAPSHOT)", and "4.0.7".
- Project Metadata:** Text input fields for "Group" (com.cognizant), "Artifact" (account), and "Package name" (com.cognizant.account).
- Packaging:** Radio buttons for "Jar" (selected) and "War".
- Configuration:** Radio buttons for "Properties" (selected) and "YAML".
- Java:** Radio buttons for "26", "25", "21", and "17" (selected).
- Dependencies:** A section with a button "ADD DEPENDENCIES... CTRL + B". It lists "Spring Web" (WEB) and "Spring Boot DevTools" (DEVELOPER TOOLS) with their descriptions.

At the bottom, there are three buttons: "GENERATE CTRL + G", "EXPLORE CTRL + SPACE", and "...".

Code:

AccountController.java

```
package com.cognizant.account.controller;

import com.cognizant.account.model.Account;
import org.springframework.web.bind.annotation.*;

@RestController
public class AccountController {

    @GetMapping("/accounts/{number}")
    public Account getAccount(@PathVariable String number) {

        return new Account(
            number,
            "savings",
            234343
        );
    }
}
```

Account.java

```
package com.cognizant.account.model;

public class Account {

    private String number;
    private String type;
    private double balance;

    public Account() {
    }

    public Account(String number, String type, double balance) {
        this.number = number;
        this.type = type;
        this.balance = balance;
    }

    public String getNumber() {
        return number;
    }

    public void setNumber(String number) {
        this.number = number;
    }

    public String getType() {
        return type;
    }

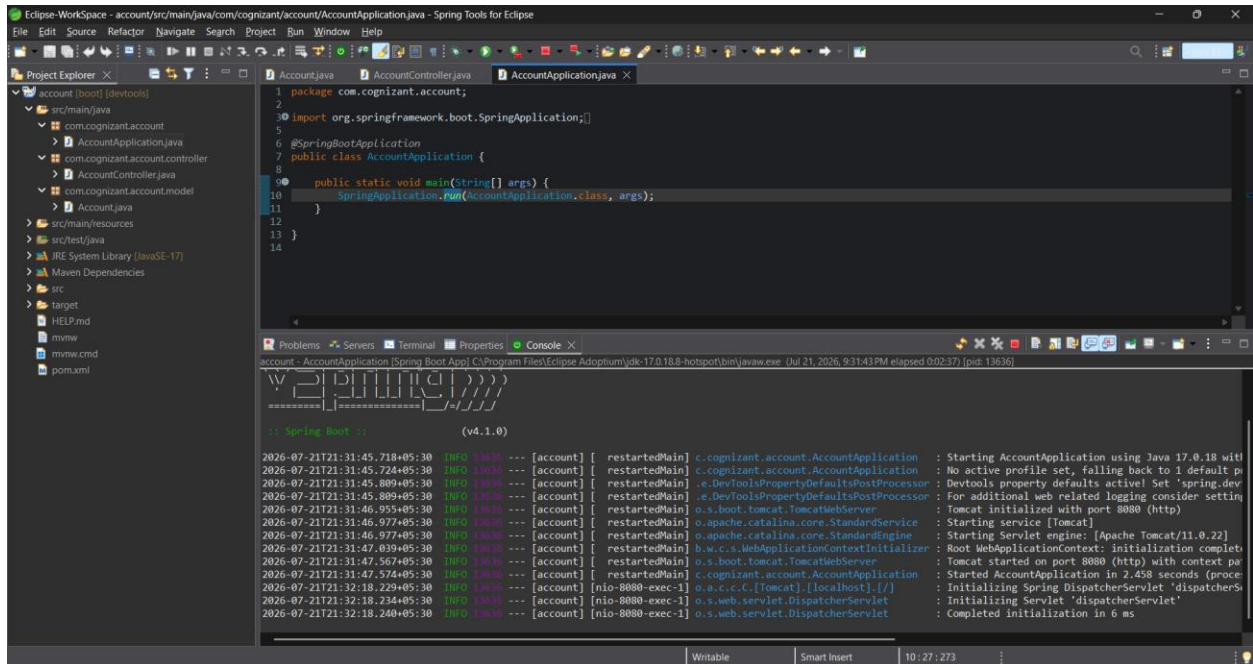
    public void setType(String type) {
        this.type = type;
    }

    public double getBalance() {
        return balance;
    }

    public void setBalance(double balance) {
        this.balance = balance;
    }
}
```

Output:

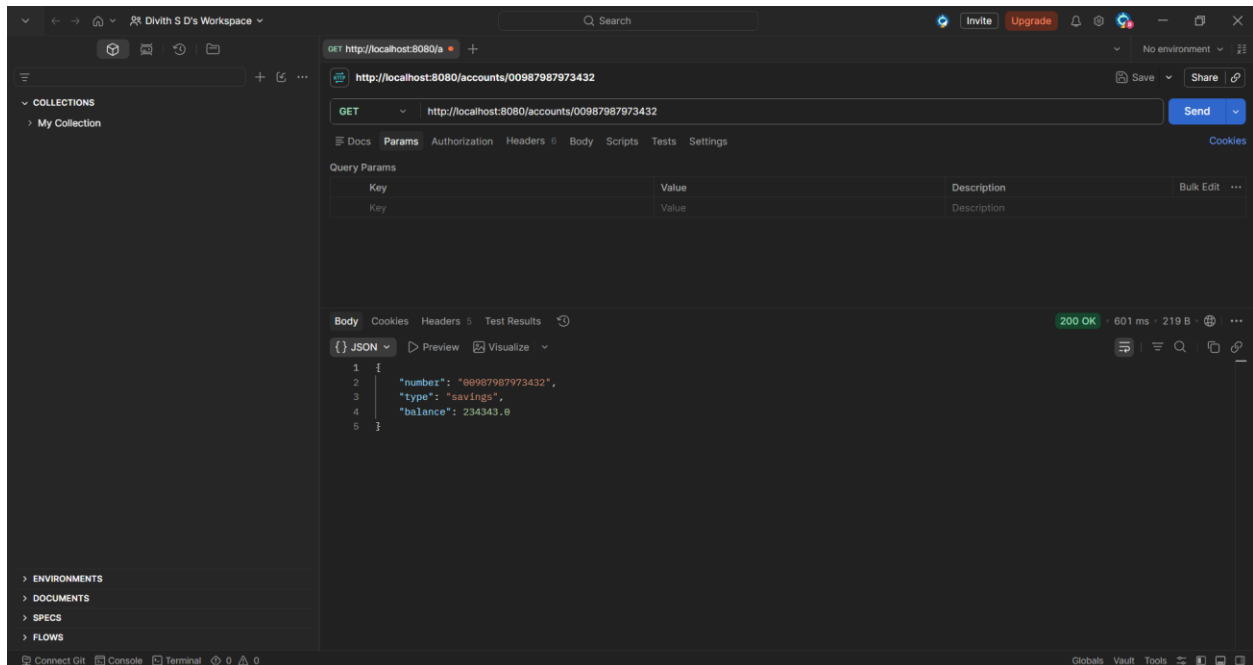
Spring Tool Compiler Output:



The screenshot shows the Eclipse IDE interface with the Spring Tool Compiler output in the console. The console output displays the following information:

```
2026-07-21T21:31:45.718+05:30 INFO 13636 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using Java 17.0.18 with
2026-07-21T21:31:45.724+05:30 INFO 13636 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default p
2026-07-21T21:31:45.809+05:30 INFO 13636 --- [account] [ restartedMain] .e.DevtoolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.dev
2026-07-21T21:31:45.809+05:30 INFO 13636 --- [account] [ restartedMain] .e.DevtoolsPropertyDefaultsPostProcessor : For additional web related logging consider settin
2026-07-21T21:31:46.955+05:30 INFO 13636 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-21T21:31:46.977+05:30 INFO 13636 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-21T21:31:46.977+05:30 INFO 13636 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-21T21:31:47.039+05:30 INFO 13636 --- [account] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization complet
2026-07-21T21:31:47.567+05:30 INFO 13636 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context pa
2026-07-21T21:31:47.574+05:30 INFO 13636 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 2.438 seconds (proce
2026-07-21T21:32:18.229+05:30 INFO 13636 --- [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherS
2026-07-21T21:32:18.234+05:30 INFO 13636 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-21T21:32:18.240+05:30 INFO 13636 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 6 ms
```

Postman API Testing:



The screenshot shows the Postman API client interface. A GET request is sent to the URL `http://localhost:8080/accounts/00987987973432`. The response is a 200 OK status with a body containing the following JSON data:

```
{
  "number": "00987987973432",
  "type": "savings",
  "balance": 234343.0
}
```

Loan Services:

Spring Initializr:

The screenshot shows the Spring Initializr web application interface. The header features the Spring logo and the text "spring initializr". On the left, there is a sidebar with a hamburger menu icon and a circular arrow icon. The main content area is divided into several sections: "Project" with radio buttons for "Gradle - Groovy", "Gradle - Kotlin", and "Maven" (selected); "Language" with radio buttons for "Java" (selected), "Kotlin", and "Groovy"; "Spring Boot" with radio buttons for "4.1.1 (SNAPSHOT)", "4.1.0" (selected), "4.0.8 (SNAPSHOT)", and "4.0.7"; "Project Metadata" with input fields for "Group" (com.cognizant), "Artifact" (loan), and "Package name" (com.cognizant.loan); "Packaging" with radio buttons for "Jar" (selected) and "War"; "Configuration" with radio buttons for "Properties" (selected) and "YAML"; and "Java" with radio buttons for "26", "25", "21", and "17" (selected). On the right, the "Dependencies" section includes a button "ADD DEPENDENCIES... CTRL + B" and two dependency cards: "Spring Web" (WEB) and "Spring Boot DevTools" (DEVELOPER TOOLS). At the bottom, there are three buttons: "GENERATE CTRL + G", "EXPLORE CTRL + SPACE", and "...".

Code:

LoanController.java

```
package com.cognizant.loan.controller;

import com.cognizant.loan.model.Loan;
import org.springframework.web.bind.annotation.*;

@RestController
public class LoanController {

    @GetMapping("/loans/{number}")
    public Loan getLoan(@PathVariable String number) {

        return new Loan(
            number,
            "car",
            400000,
            3258,
            18
        );
    }
}
```

Loan.java

```
package com.cognizant.loan.model;

public class Loan {

    private String number;
    private String type;
    private double loan;
    private double emi;
    private int tenure;

    public Loan() {
    }

    public Loan(String number, String type,
                double loan, double emi,
                int tenure) {

        this.number = number;
        this.type = type;
        this.loan = loan;
        this.emi = emi;
        this.tenure = tenure;
    }

    public String getNumber() {
        return number;
    }

    public void setNumber(String number) {
        this.number = number;
    }

    public String getType() {
        return type;
    }

    public void setType(String type) {
        this.type = type;
    }

    public double getLoan() {
        return loan;
    }

    public void setLoan(double loan) {
        this.loan = loan;
    }

    public double getEmi() {
        return emi;
    }
}
```

```

    public void setEmi(double emi) {
        this.emi = emi;
    }

    public int getTenure() {
        return tenure;
    }

    public void setTenure(int tenure) {
        this.tenure = tenure;
    }
}

```

Output:

Spring Tool Compiler Output:

```

Eclipse-WorkSpace - loan/src/main/java/com/cognizant/loan/model/Loan.java - Spring Tools for Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
  account [boot] [devtools]
  loan [boot] [devtools]
    src/main/java
      com.cognizant.loan
        LoanApplication.java
        com.cognizant.loan.controller
        LoanController.java
        com.cognizant.loan.model
        Loan.java
      src/main/resources
      src/test/java
      JRE System Library (JavaSE-17)
      Maven Dependencies
      src
      target
      HELP.md
      mvnw
      mvnw.cmd
      pom.xml

Account.java AccountController.java AccountApplication.java LoanController.java Loan.java LoanApplication.java

49 public double getEmi() {
50     return emi;
51 }
52
53 public void setEmi(double emi) {
54     this.emi = emi;
55 }
56
57
58 public int getTenure() {
59     return tenure;
60 }
61
62 public void setTenure(int tenure) {
63     this.tenure = tenure;
64 }
65 }

Problems Servers Terminal Properties Console
terminated> loan - LoanApplication [Spring Boot App] C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot\bin\java.exe (Jul 21, 2026, 9:36:59 PM - 9:37:02 PM elapsed 0:00:02) [pid: 23488]

Spring Boot (v4.1.0)

2026-07-21T21:37:01.259+05:30 INFO 23488 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 17.0.18 with PID :
2026-07-21T21:37:01.263+05:30 INFO 23488 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to 1 default prof
2026-07-21T21:37:01.311+05:30 INFO 23488 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtoo
2026-07-21T21:37:01.311+05:30 INFO 23488 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting tl
2026-07-21T21:37:01.995+05:30 INFO 23488 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-21T21:37:02.009+05:30 INFO 23488 --- [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-21T21:37:02.009+05:30 INFO 23488 --- [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-21T21:37:02.056+05:30 INFO 23488 --- [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed
2026-07-21T21:37:02.364+05:30 WARN 23488 --- [loan] [ restartedMain] ConfigServletWebServerApplicationContext : Exception encountered during context initialization
2026-07-21T21:37:02.377+05:30 INFO 23488 --- [loan] [ restartedMain] .s.b.a.l.ConditionEvaluationReportLogger :

Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.
2026-07-21T21:37:02.390+05:30 ERROR 23488 --- [loan] [ restartedMain] o.s.b.d.logging.FailureAnalysisReporter :

*****
APPLICATION FAILED TO START
*****

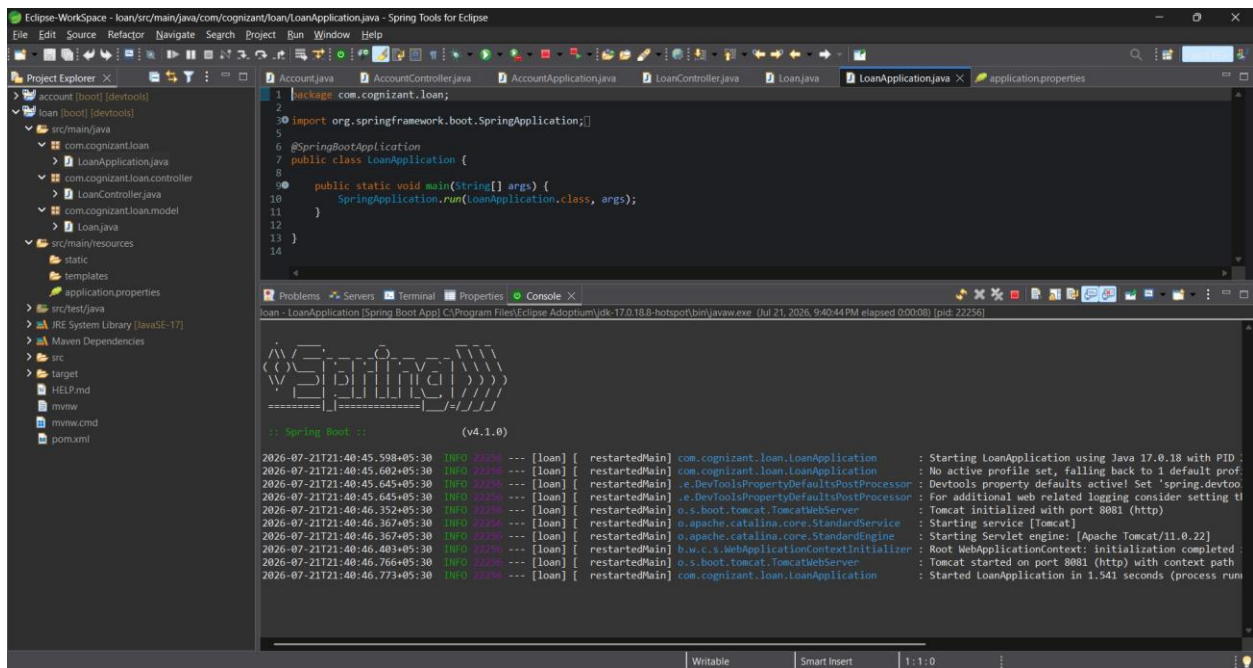
Writeable Smart Insert 57:1:1050

```

Failed due to the already account services is using 8080 port.

Updating application.properties:

With server.port = 8081



Postman API Testing Output:

